

Streamline Simplification — Merging & Splitting

Angle criterion (Xiao Eq. 9)

$$120^\circ \leq \alpha \leq 240^\circ \text{ (3-valent)}, \quad 144^\circ \leq \alpha \leq 216^\circ \text{ (5-valent)}$$

Blend formula (Xiao Eq. 10 / Kowalski Eq. 28)

$$\gamma_{12}(s) = (1 - s) \gamma_1(s) + s \gamma_2(s)$$

- Merge nearby streamlines at singularities (angle criterion)
- Split at transversal crossings (T-junctions)
- Result: clean quad-only block boundaries

Code: `StreamlineMerging`, `StreamlineIntersectionSplitter` *Lit:* Xiao et al. 2020,
“Robust topology decomposition of 2D domains” (Alg. 2)